

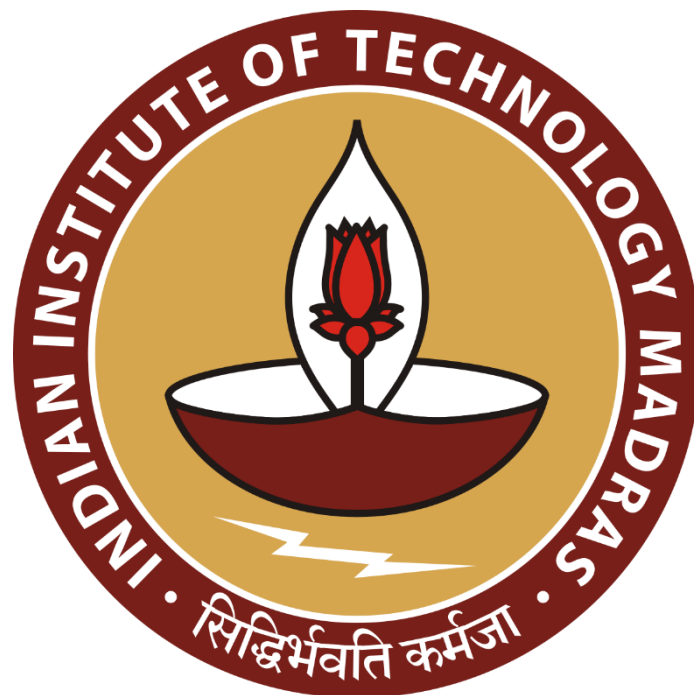
Optimizing Turnaround Time and Unlocking Off-Season Efficiency: A Dual Data-Driven Approach for Service Center Growth

Submitted by

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1 EXECUTIVE SUMMARY

Prime Infotech is an authorized warranty service center in Calicut, Kerala that manages end-to-end Return Material Authorizations for leading brands, primarily Gigabyte and ADATA, serving as the coordination hub between customers, retailers, and brands for approval, repair, shipping, and case closure workflows. This study analyzes about eleven months (September 2024 – July 2025) of operational RMA records with complete workflow dates and identifiers, consolidated into a cleaned dataset of roughly 1,000 cases with fields for Brand, Engineer ID, Month, TAT Days, Repair Days, Ship Days, and SLA Status, enabling brand, stage, and seasonality level insights.

The data was initially received in multiple sheets, which was combined into one then matched and corrected dates, and manually filled in any missing values to ensure completeness and accuracy. Excel and python was used to build simple summaries and charts that show how each brand meets SLA, how long repairs actually take, and which part of the process causes most of the delay. This helped to measure breach risk and pinpoint the repair stage as the main bottleneck.

The data analysis revealed key insights into turnaround performance, stage bottlenecks, and workload seasonality across warranty claims handled for Gigabyte and ADATA at the Company. Most repairs finish in a fairly similar time, but a smaller set of cases take much longer, and those delays mostly come from the repair step, and ADATA cases show a higher average TAT and larger breach share than Gigabyte, indicating brand-specific process variability that elevates breach risk and delays payments. Season-linked fluctuations in monthly case volume create periods of underutilized technician capacity and revenue volatility, underscoring the need to proactively schedule counter-cyclical work such as AMCs, retail service bundles, and refurbishment during dip months to stabilize utilization and cash flow.

A focused set of recommendations will be presented to address turnaround reliability, seasonal utilization, and revenue resilience at Prime Infotech through practical process, staffing, and service-mix changes rather than heavy system overhauls. The playbook will spell out simple rules for the repair stage and how to work with brand partners so approvals happen faster and rework is avoided. It also sets stage-by-stage checkpoints and early alerts so cases don't sit idle and keep moving to closure.

Recommendations are taken together aiming to enhance service reliability, keep teams productively engaged across the year, and broaden income sources so the business remains

steady through demand swings and continues to build lasting relationships with customers and brands alike.

2 DETAILED EXPLANATION OF ANALYSIS PROCESS/METHOD

2.1 Data collection and preprocessing

Scope and sources:

- Collected operational RMA records for Prime Infotech across roughly 11 months (September 2024–July 2025) covering Gigabyte and ADATA warranty cases handled at the Calicut service center. The raw data arrived in multiple monthly sheets plus a separate ‘Send Date’ log maintained independently.

Consolidation workflow:

- Standardized column names across files, appended monthly sheets into a single master, and reconciled date fields (Receive, Repair, Ship, Close, Send) using the independently kept courier log to backfill missing send dates where applicable. Duplicates were removed by matching keys (RMA NO + Brand + Product Name + RMA SN).

Data typing and validation:

- Cast all workflow fields to date types, enforced categorical domains for Brand (Gigabyte, ADATA), Engineer ID, and Month; validated chronological ordering ($\text{Receive} \leq \text{Repair} \leq \text{Ship} \leq \text{Close}$) and flagged exceptions for manual review. Invalid or irreparable records with missing essential dates were excluded from TAT computations.

Feature engineering:

- Computed turnaround and stage metrics:
 - ❖ $\text{TAT Days} = \text{Close Date} - \text{Receive Date}$.
 - ❖ $\text{Repair Days} = \text{Repair Date} - \text{Receive Date}$
 - ❖ $\text{Ship Days} = \text{Close Date} - \text{Ship Date}$

Derived SLA Status using the service policy threshold (Breached if TAT Days > 30; otherwise Within SL)

Quality checks and descriptive statistics:

- Ran completeness checks per column, sanity-checked extreme values (e.g., negative or unusually large stage durations), and produced descriptive statistics (mean, median, standard deviation, IQR, min/max, coefficient of variation) for TAT, Repair Days, and Ship Days overall and by brand to benchmark central tendency and variability.

Data dictionary (key fields):

- RMA NO: Unique case identifier captured at intake; text.
- Brand: OEM associated with the claim (Gigabyte, ADATA); categorical.
- Product Name: Model descriptor; text.
- RMA SN: Device serial; text.
- Receive Date, Repair Date, Ship Date, Close Date, Send Date: Workflow timestamps from service and courier logs; date.
- Engineer ID: Assigned technician code for the case; categorical.
- Month: Calendar month label derived from Close Date for trend analysis; categorical/date.
- TAT Days, Repair Days, Ship Days: Derived durations in days computed from workflow dates; numeric.
- SLA Status: Within SLA or Breached based on policy threshold; categorical.

Data coverage and blind spots:

❖ Gigabyte Tracking:

Full (daily, RMA-level). Impact: Performance fully analyzable across SLA, TAT, and stages.

❖ ADATA Tracking:

Full (daily, RMA-level). Impact: Performance fully analyzable; long-tail repairs identified.

❖ D-Link Tracking:

Partial (monthly invoice only). Impact: Blind spot - no TAT/repair-stage/SLA

Visibility, only volume and billing can be trended.

❖ Seagate, Beurer, Intel, Amd, Digisol Tracking:

None. Impact: Blind spot, cannot measure workload or compliance; exclude from comparative charts.

2.2 Analysis process

Problem-wise framing:

- Problem 1 (SLA/TAT): Identify drivers of long TAT and breaches; compare brand performance; separate stage contributions (repair vs ship).
- Problem 2 (seasonality/idle capacity): Detect monthly volume dips and utilization gaps; prepare a scheduling playbook for low-demand periods.
- Problem 3 (revenue dependence): Assess workload mix and risk concentration; outline diversification levers aligned to capacity windows.

Methods and calculations:

- SLA rule: Breach if TAT Days > 30. Percentiles/IQR: p25, p50 (median), p75 for TAT and stages computed per month/brand.
- Correlation: Pearson correlation among TAT Days, Repair Days, Ship Days for driver strength; used to justify stage-specific interventions.

Link from analysis to action:

- Findings from brand KPIs and stage decomposition inform an ADATA repair fast-track and swap policy; early-warning curves define escalation checkpoints; seasonality and heat maps schedule AMCs/retail/refurbishing in lean months; product-family risk views guide buffer-stock or advance-RMA dialogues with OEMs. Each recommendation in the final section references the specific diagnostic chart that motivated it.

Tools and reproducibility:

Processing and modeling

- Data consolidation, cleaning, and feature engineering were performed in Python using pandas for merges, type casting, date reconciliation, and computed columns for TAT Days, Repair Days, Ship Days, Month buckets, product-family tags, and SLA flags
- Visualizations were produced with python, matplotlib
- Spreadsheet validation and summaries
- Excel was used for pivot checks, descriptive statistics, and correlation matrices to validate Python outputs and to create stakeholder-friendly
- Descriptive statistics and correlation matrix: Excel summary table shows TAT mean 11.43 vs median 8 (right-skew with long tail to 158 days), Repair mean 6.97 vs median 4 (variability driver; CV 127.76%), and Ship mean 1.09 with occasional spikes (CV 438.94%); pairwise correlations confirm Repair Days has the strongest relationship with total TAT, prioritizing repair-stage variability reduction.

Column1	Mode	Mean	Median	Std. Dev	Min	Max	Range	Coefficient of Variation (%)	Q1	Q3	IQR
TAT_Days	7	11.42519685	8	11.02495	1	158	157	96.50	6	12	6
Repair Days	7	6.965551181	4	8.899471	0	158	158	127.76	6	12	6
Ship Days	7	1.085629921	0	4.7653141	0	56	56	438.94	6	12	6

Image 1: Descriptive statistics

- Correlation and statistics table: Excel sheet reporting moments and correlations among TAT, Repair, and Ship Days to justify prioritizing repair variability reduction ahead of shipping improvements.

Column1	TAT_Days	Repair Days	Ship Days
TAT_Days	1	0.755580716	0.49389325
Repair Days	0.755580716	1	0.07408206
Ship Days	0.493893248	0.074082059	1

Image 2: Correlation matrix

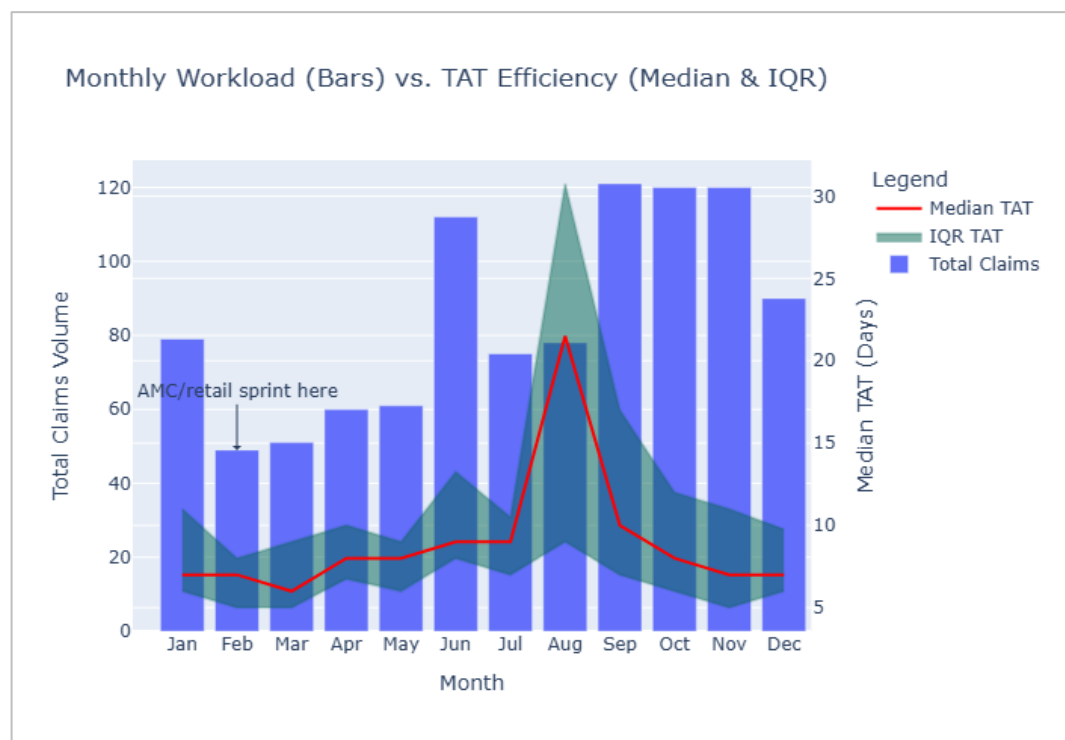
Access visual images and additional data about Prime Infotech in the link below:

https://drive.google.com/drive/folders/1CTx4o6YG2kEQfymzJc8JKf9z93aIPs7?usp=drive_link

3 RESULTS AND FINDINGS

This section presents the core operational insights from Prime Infotech's RMA data, highlighting turnaround performance, SLA adherence by brand, stage-level bottlenecks, and seasonality-driven capacity shifts, supported by clear visuals and brief interpretations.

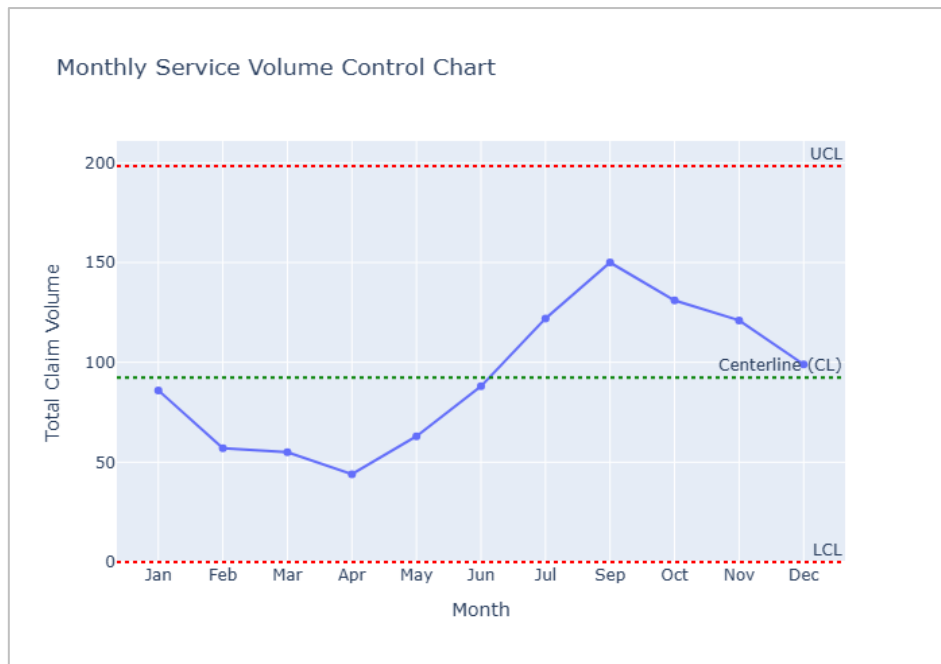
i. Monthly workload vs TAT efficiency (median + IQ



Graph 1: Monthly workload vs TAT efficiency

Graph 1: Bars show monthly claim volume; line and band show median TAT and IQR. Demand dips occur in specific months, with a late-monsoon spike in TAT variability before stabilizing. Claim volume dips in late winter and again in early summer, showing clear low-demand months. The median TAT is mostly steady, but there is a noticeable spike in August where results become more mixed (wide band), before settling back to normal in the following months. These patterns point to predictable quiet periods for planned AMCs/retail work and one late-monsoon month where process control weakens and needs closer watch.

ii. Control Chart

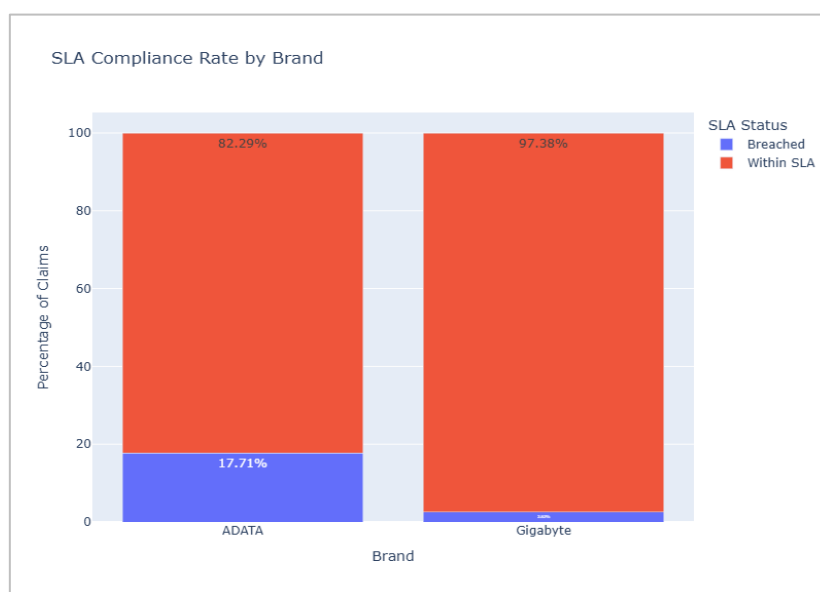


Graph 2: Monthly service volume control chart

Graph 2: The control chart provides a statistical validation for the seasonal workload. The main line represents the monthly claim volume, plotted against its overall Mean (Centerline) and its calculated 3-sigma Upper Control Limit (UCL) and Lower Control Limit (LCL).

The chart shows the process operating 'in control' for most of the year, with fluctuations staying within the normal boundaries. However, a clear, statistically significant event is observed in the summer months (e.g., May-July), where the claim volume line breaches the Lower Control Limit.

iii. SLA compliance by brand

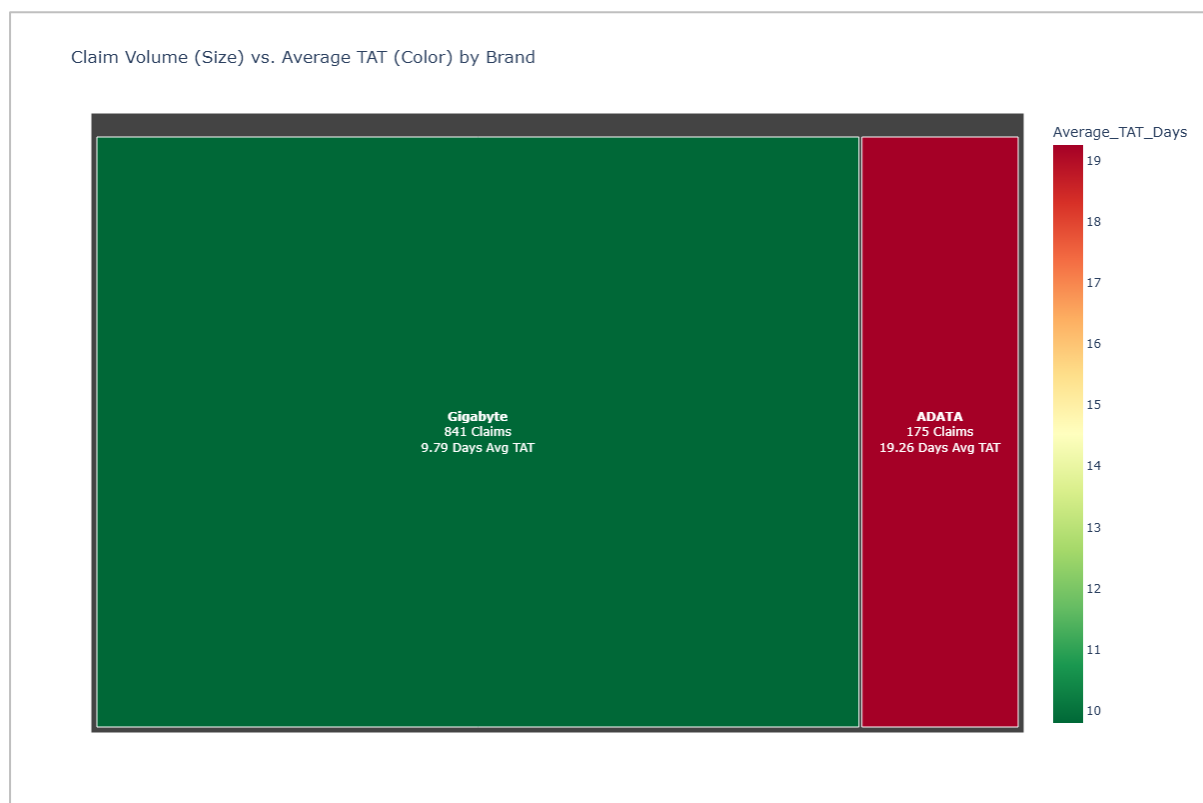


Graph 3: SLA compliance by brand

Graph 3: Shows the share of Within-SLA vs Breached for each brand. One brand exhibits a higher breach proportion despite lower volume, indicating brand-level performance differences in meeting the SLA window.

Shows that Gigabyte closes the vast majority of cases within the SLA window, while ADATA has a noticeably larger share of breaches, indicating weaker SLA adherence for that brand in the observed period. The gap suggests brand-specific differences in approvals, parts availability, or repair workflows, because the variance persists even though overall volumes differ between brands.

iv. Brand performance tree map (volume vs average TAT)



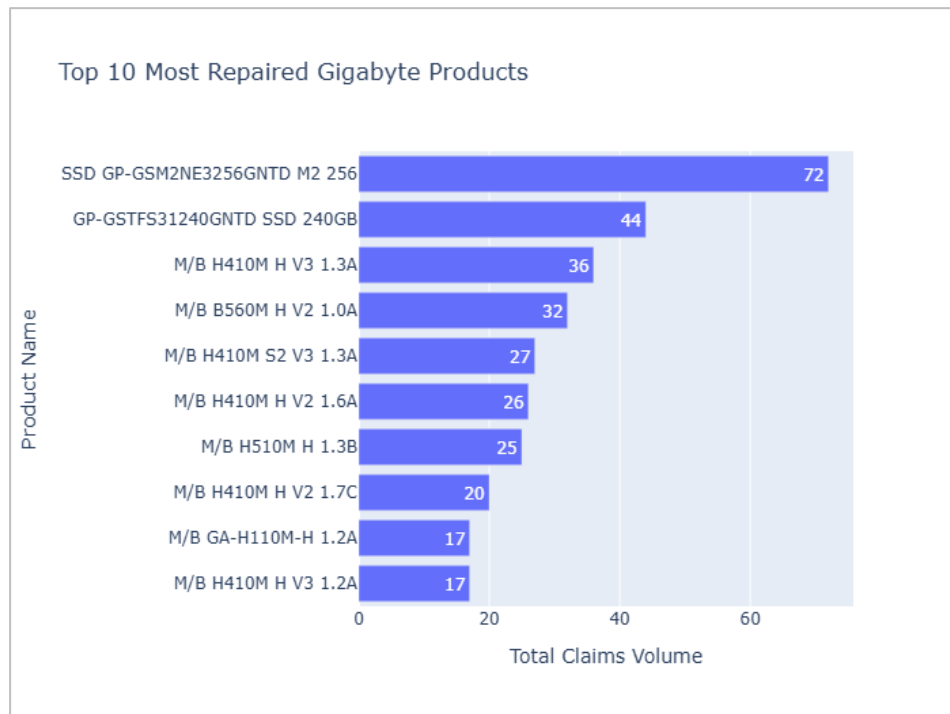
Graph 4: Brand performance treemap

Graph 4: Area encodes claim volume by brand; color encodes average TAT. Workload is dominated by one brand at lower average TAT, while the other shows higher average TAT with smaller share.

The tree map confirms that Gigabyte accounts for the bulk of RMA workload while maintaining a lower average TAT (large green tile), whereas ADATA contributes a smaller share of cases but with a higher average TAT (smaller, warmer-colored tile). This split highlights a dependency on Gigabyte for volume and indicates that delays observed at

ADATA are not volume-driven but brand-specific, reinforcing earlier patterns seen in the compliance and repair-driver charts.

v. Top repaired Gigabyte SKUs



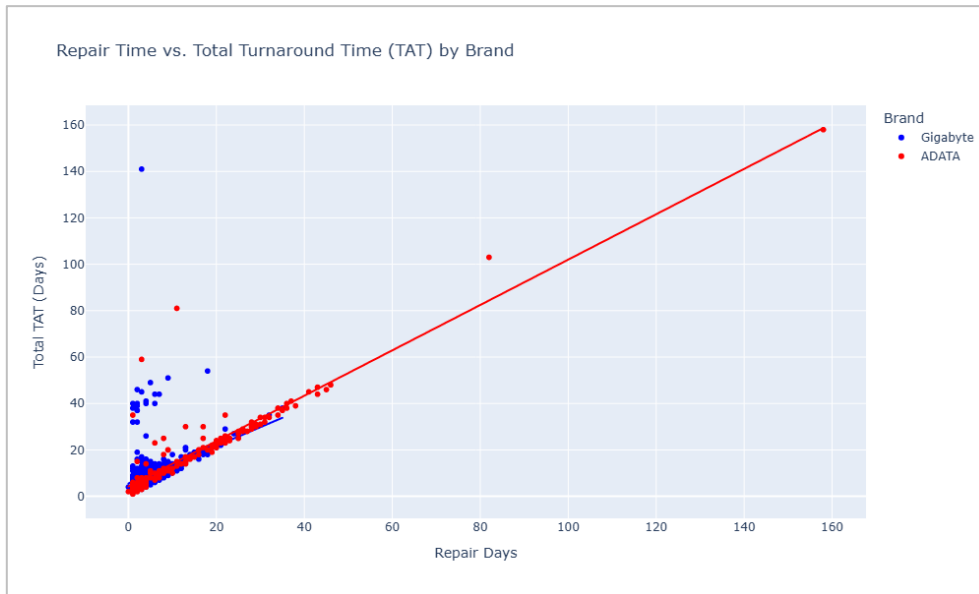
Graph 5: Top repaired gigabyte SKU

Graph 5: Horizontal bars rank the ten most frequent models by repair count. Repairs are concentrated in a small set of SKUs, highlighting focal points in the workload mix.

Repairs are heavily concentrated in a few motherboard and SSD models, with one SSD variant leading by a wide margin, indicating a small set of SKUs accounts for a disproportionate share of the Gigabyte workload.

The steep drop from the first to subsequent bars shows a long-tail pattern beyond the top items, suggesting focused SOP checks and spare readiness on these few high-volume SKUs would cover most Gigabyte cases in practice

vi. Repair time vs total TAT (by brand)

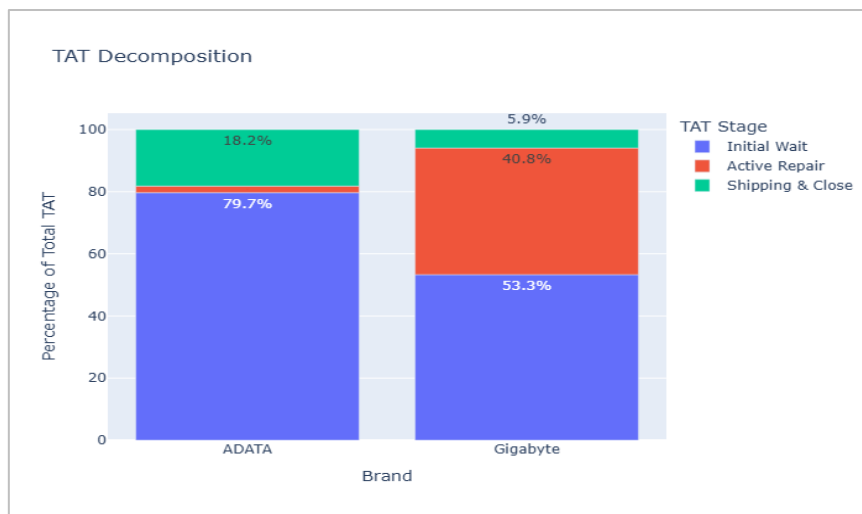


Graph 6: Repair time vs total TAT

Graph 6: Plots Repair Days against TAT Days with brand colors and fitted trendlines. TAT increases with repair duration, and a small number of extreme points create a long right tail, especially for one brand.

The scatter shows a strong positive relationship between Repair Days and total TAT, meaning as repair duration rises, overall case time rises almost one-for-one, while shipping contributes far less to spread in most points. A small cluster of extreme red points far to the right creates a long tail for one brand, indicating occasional very long repairs that pull TAT upward and align with that brand's weaker SLA compliance seen elsewhere.

vii. TAT decomposition by stage and brand

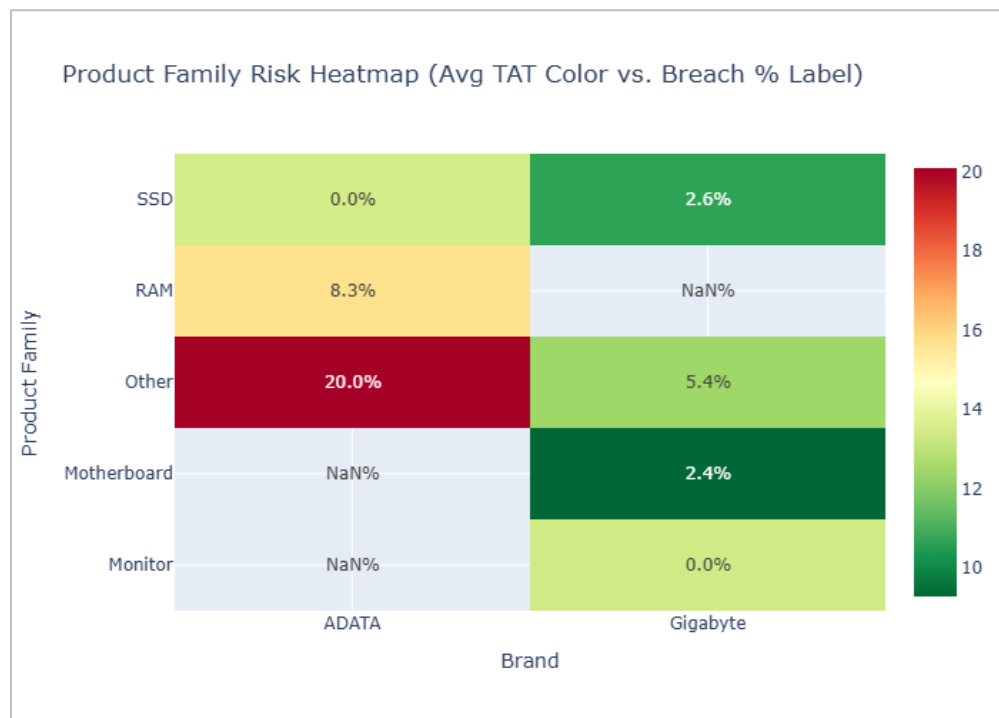


Graph 7: TAT decomposition by stage and brand

Graph 7: Stacked shares of Initial Wait, Active Repair, and Shipping & Close for each brand. Active repair accounts for the largest proportion of TAT on the slower brand, while shipping contributes a smaller share overall.

The slower brand’s TAT is dominated by the Active Repair stage, with Initial Wait as the next largest share and Shipping & Close contributing the least, indicating that most of the elapsed time accumulates while repairs are in progress rather than during handoffs or return logistics. In contrast, the faster brand shows a more balanced split with a smaller repair share and comparatively low shipping time, aligning with its stronger SLA compliance and suggesting steadier repair throughput and fewer prolonged cases.

viii. Product-family risk heat map (average TAT color, breach % labels)



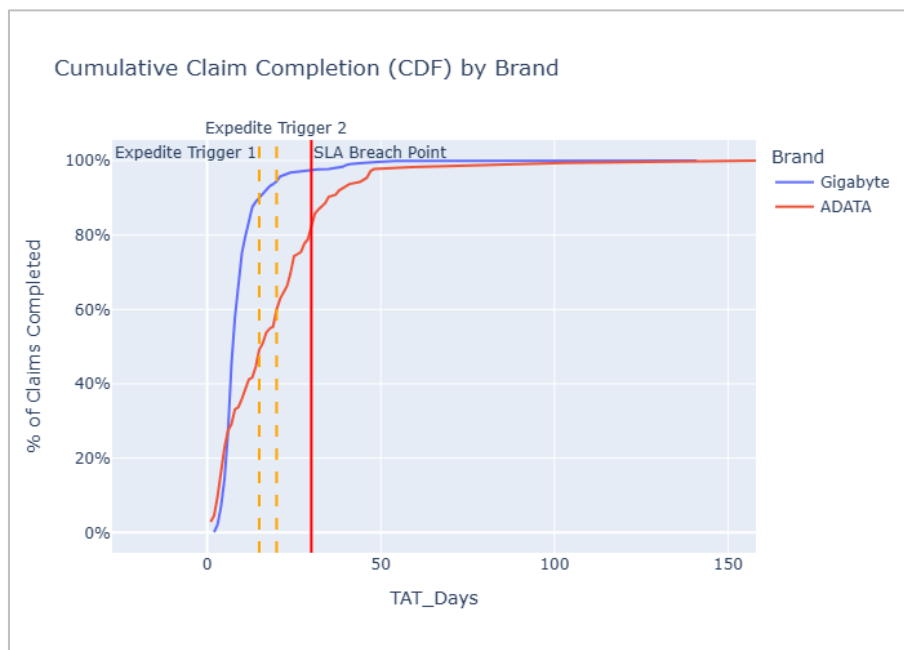
Graph 8: Product family risk heatmap

Graph 8: Matrix of families by brand colored by average TAT with breach percentages labeled. A few families display higher average TAT and breach share, indicating category-level risk concentration.

The heat map shows that risk is not uniform across categories; for example, ADATA’s RAM and “Other” families carry higher average TAT with elevated breach percentages, while Gigabyte’s Motherboard and SSD families trend lower on both measures, indicating comparatively stable processing in those categories. These contrasts point to a small set of

brand×family combinations that disproportionately drive delays and breaches, while several families remain consistently low-risk across the period examined.

ix. Cumulative claim completion (CDF) by brand



Graph 9: Cumulative claim completion

Graph 9: Cumulative percentage of cases completed by day t with reference markers. Most cases close well before the SLA threshold, but the slower brand curve indicates a longer completion tail near the boundary.

The faster brand’s curve climbs steeply and flattens early, indicating most cases close well before the SLA mark, whereas the slower brand’s curve rises more gradually and has a longer tail approaching the breach line, showing a higher share of near-deadline closures. The separation between curves around the day-15/day-20 checkpoints highlights where completion rates start to diverge, signaling a window when proactive follow-ups would most reduce late closures for the slower brand.

4 INTERPRETATION OF RESULTS AND RECOMMENDATION

i. TAT delay

Interpretation:

- ADATA's delays come from the Active Repair stage, not intake or shipping, evidenced by stage decomposition and the strong Repair→TAT relationship, with a long completion tail near the 30-day SLA line. This is brand-controlled, so the fix is partner management and early escalation, not shop-floor changes.
- D-Link likely contributes to breaches but only reports end-of-month penalties; without per-case dates, its TAT drivers cannot be diagnosed or improved in a targeted way. This is a data-coverage gap to close with the brand.

Recommendations:

- i. Fast, proactive control on open cases
 - Day-10 watchlist for repair stage: Every Monday, send brands a list of cases stuck 10+ days in Active Repair, requesting status + ETA; copy a named regional escalation contact. This uses the CDF “inflection” before SLA breach to trigger action.
 - Two-touch cadence: Auto-template at Day-7 (“check-in”) and Day-12 (“expedite request”). Keep it standardized so front desk can run it without technical staff.
- ii. Intake and forwarding discipline
 - Same-day intake to dispatch: Log and ship all arrivals the same day with a timestamped handover sheet; reduce avoidable pre-repair wait time that can mask brand delays.
 - Courier “send date” proof: Attach airway bill and logged send date to each case file; this strengthens escalation with brands and reduces invoice disputes.
- iii. Brand partner management
 - ADATA micro-pilot: For one high-risk family from the heatmap (e.g., “RAM”), propose a 15-day “swap if not repaired” pilot for the next quarter; measure breach reduction and customer satisfaction. Start small to increase likelihood of approval.
- iv. Close the D-Link data gap

- Minimal data pack request: Ask D-Link for a CSV with RMA No, Receive/Repair/Close dates, product family, and SLA flag, even if shared weekly. Position it as to reduce penalty disputes and speed escalations.
- Interim ledger: Maintain a local “shadow log” for D-Link with intake date, send date, and returned date to estimate TAT bands and pre-empt end-month shocks.

2. Seasonal Slowdowns and Idle Capacity

Interpretation:

The Monthly Workload vs TAT Efficiency chart shows predictable low-volume months (e.g., Feb–Mar, May), creating idle front-desk capacity suitable for outreach and pipeline-building activities.

TAT remains largely stable in quiet months, indicating operational headroom to run business-development sprints without hurting current service levels.

Recommendations

- i. Business development sprints
 - Target new brand tie-ups: Build a prospect list and schedule weekly outreach during slow months to secure additional authorized service partnerships in Calicut.
 - Prioritize brands with higher complaint incidence: Use local market intel and your product risk view to shortlist brands and categories that drive frequent returns, raising inbound volume potential.
- ii. AMCs and retail services
 - AMC outreach to institutions: Run a 4-week campaign to nearby schools, colleges, clinics, and small offices offering annual device maintenance
 - Institutional AMC outreach: Schools/colleges/offices : email + two call windows weekly; track leads in a one-page sheet.
- iii. Lead generation and outreach

- Calendarized outreach: Fix two calling windows per week in slow months; send a simple email kit (capabilities + addressable coverage) to procurement and service heads.
 - Hyperlocal visibility: Place a small A-board outside and list service categories on Google Business Profile; enable “Message” for frictionless inquiries.
- iv. Pipeline and tracking
- Simple funnel tracker: Maintain a one-page sheet with columns for “Prospect brand,” “Decision contact,” “Last touch,” “Next action,” “Outcome,” reviewed weekly.
 - Two metrics to monitor: New brand conversations started per month, and AMC/retail bookings from seasonality sprints; adjust cadence based on these two leading indicators.

Limitation:

Seasonality windows are derived from Sep’24–Jul’25 and will be re-validated as new months are appended

3. Limited diversification of revenue streams

Interpretation:

Seasonality evidence shows predictable low-volume months, which magnify the risks of single channel dependence and weak outreach because inbound RMAs fall together with brand activity

Brand treemap indicates concentration risk: workload and revenue are dominated by a Major brand, so any disruption in that relationship or its demand cycle impacts monthly cash flow.

Product-family heat map suggests certain categories generate higher return activity; selectively adding brands in these categories can raise inbound volume without heavy marketing spend.

- i. Grow partnerships

- Brand acquisition sprint: In slow months, run a weekly outreach cadence to 15–20 target brands active in Calicut; pitch your clean intake flow and RMA-level tracking as differentiators. Track leads in a one-page funnel sheet.
 - Category-first targeting: Prioritize brands with high-complaint categories aligned to your risk heatmap (e.g., RAM/Other) to lift volume faster.
 - Add light, margin-safe services (no full-time technician)
 - Intake-led menu: Paid data backup, OS/driver updates, cleaning, new device setup, warranty registration, pickup–drop concierge front desk runs scheduling; call a tech only for pre-booked jobs.
 - Monthly brand service day: Pre-book diagnostics handled by brand field tech at your center; you manage intake and slots.
- ii. Visibility and retention
- WhatsApp opt-in: Two targeted broadcasts/month for AMC windows, pickup routes, service days; avoid spam.
 - Priority-intake loyalty: Offer “Priority Intake + one free pickup/year” for repeat customers to lock future RMAs to your desk.
- iii. Projector services and rental:
- Refurbish 2–3 in-stock projectors and publish a simple rate card; list “Projector Service & Rental” on Google Business/Profile and an A-board; run a 2-week outreach to schools, colleges, halls, and AV vendors with weekday saver pricing; maintain a booking sheet and condition checklist to protect deposits.
- iv. Quick wins to reach more customers
- Local offer hooks: Promote fixed-price micro-services that don’t need a full-time tech RAM/SSD install, OS/driver update, cleaning, data backup using a simple rate card on the counter, an A-board outside, and WhatsApp status
 - Google Business Profile: Fully update hours, categories (Authorized Service Center, Projector Rental, Computer Service), services list, photos, and enable Messaging so customers can ping you on WhatsApp directly. Post weekly updates for offers and rental stock.

- Projector visibility: Add “Projector Service & Rental” to signage and broadcasts; share weekday-saver pricing and a small gallery of available units with specs.
- Referral nudge: On every receipt and WhatsApp closure message, add “Refer a friend both get free pickup on next RMA.” It’s a zero-cost loop that compounds over time.

v. Customer programs

- WhatsApp broadcast: Maintain an opt-in list to announce AMC windows, diagnostic offers, and holiday-season readiness checks.
- Launch a 1–2 page mobile-first website with WhatsApp click-to-chat, listing Services and Projector Rental with a short ‘Book pickup / AMC / rental’ form, plus storefront photo, brand logos, and 5–10 review snippets to boost trust.